

Native multi-view 3D with AnySplat

Upload overlapping photographs, reconstruct one Gaussian scene on an NVIDIA CUDA workstation, and inspect it from every direction in the local WebGL viewer.

MODEL	INPUT	COMPUTE
AnySplat	2-16 uncalibrated views	NVIDIA CUDA
Depth Anything V2	1 photograph	CPU or CUDA

1. Prepare the workstation

Use 64-bit Python 3.11. Place the project at D:\GaussianSceneStudio. Do not copy a .venv from another computer.

Double-click **Setup NVIDIA Workstation.cmd**. The first setup downloads CUDA-enabled PyTorch, the pinned 2.94 GB AnySplat checkpoint, and the small depth fallback. It verifies the checkpoint and performs a real CUDA tensor calculation.

```
cd D:\GaussianSceneStudio
powershell -NoProfile -ExecutionPolicy Bypass -File .\setup.ps1 -Device CUDA
.\.venv\Scripts\python.exe scripts\doctor.py --require-cuda --require-anysplat
```

The screenshot supplied for the workstation shows an RTX A1000 with about 8 GB VRAM. Close QGIS and other GPU-heavy programs before reconstruction.

Give the model useful overlap

AnySplat estimates cameras and a single scene jointly. Image quality and overlap matter more than raw image count.

1

Keep the subject still

Move the camera around one object or room. Do not move the subject between frames.

2

Follow a camera path

Upload neighboring views in order. Each view should share visible features with the next.

3

Keep framing consistent

Use sharp images, similar zoom, stable orientation, and consistent lighting.

4

Start with Auto-safe

On the 8 GB RTX A1000, Auto-safe selects two evenly spaced views. Try four only after two-view inference succeeds with free VRAM.

5

Inspect and export

Drag to orbit, right-drag to pan, scroll to zoom, then export the full .ply or save a viewport image.

View budget

Auto-safe uses 2 views at 8 GB, 4 at 12 GB, 6 at 16 GB, 10 at 24 GB, and up to 16 above 24 GB. These are conservative heuristics, not guarantees. Use every upload can require much more VRAM.

Know what passed and what remains

The CPU computer can validate application logic, formats, and the viewer. Only the NVIDIA workstation can validate a real AnySplat forward pass.

Validated locally

API and upload limits; ordered 2-16 image jobs; engine contracts; cancellation and recovery; PLY round trips; AnySplat tensor conversion; bounded outputs; deterministic view sampling; WebGL asset delivery; JavaScript syntax; pinned source identity.

Workstation acceptance

1. Setup and doctor commands exit cleanly.
2. Calibration scene orbits and exports.
3. Two overlapping views complete with method anysplat and device cuda in scene details.
4. Exported PLY reloads and displays correctly.
5. Cancellation releases the worker slot.
6. Four-view test runs only if GPU memory remains safe.

Limits

Unseen regions may remain incomplete. Inputs are centre-cropped to 448 x 448. The portable export keeps at most two million strongest Gaussians; the interactive preview keeps at most 1.5 million. CUDA out-of-memory errors do not silently substitute another model.

Licence note

The AnySplat repository and published model state MIT. The upstream source also contains individual third-party non-commercial headers. Preserve notices and obtain organisational licence review before claiming the full dependency bundle is cleared for commercial redistribution. See docs/MODEL-LICENSES.md.

Official sources

github.com/OpenRobotLab/AnySplat
huggingface.co/lhjiang/anysplat
arxiv.org/abs/2505.23716